ThyroTrack: Holistic Endocrinal Cancer Prognosis Platform

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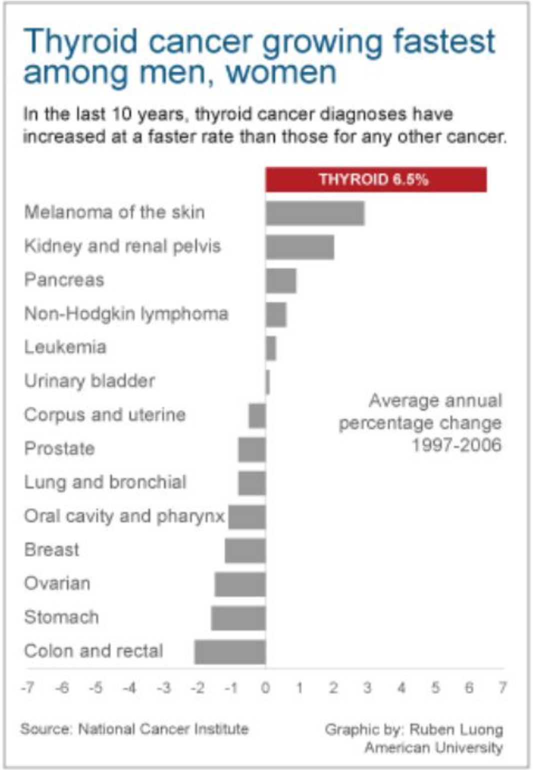
Background Information

- 67% of adults have thyroid nodules; only 5–15% are malignant.
- Overdiagnosis → unnecessary thyroidectomies, cost, complications.
- Bethesda system classifies thyroid biopsy cytology into 6 diagnostic categories.

Figure 1

Thyroid Cancer Incidence As Compared to Other Cancers



Note. Bryde et al., 2016

Background Information

- Existing AI models:
  - Mostly binary classifiers
  - Rarely model progression
  - Don't incorporate patient similarity
- This leads to misclassification in the "gray zone" (Bethesda III–IV).

Figure 1

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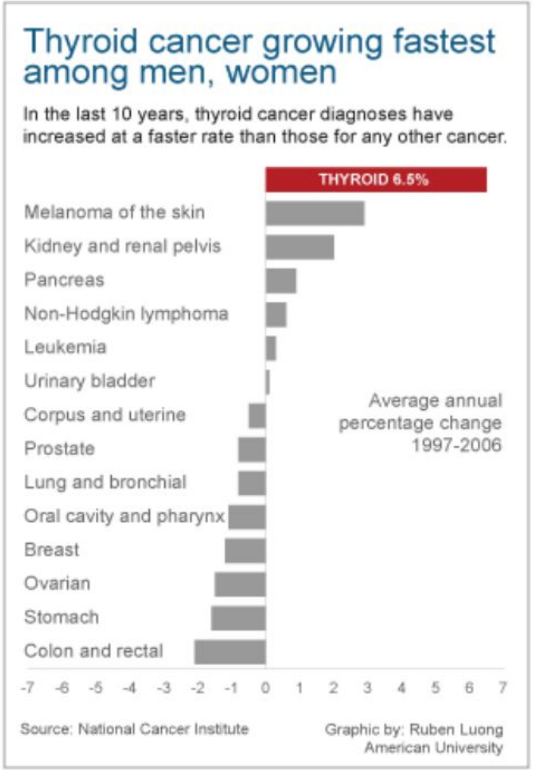

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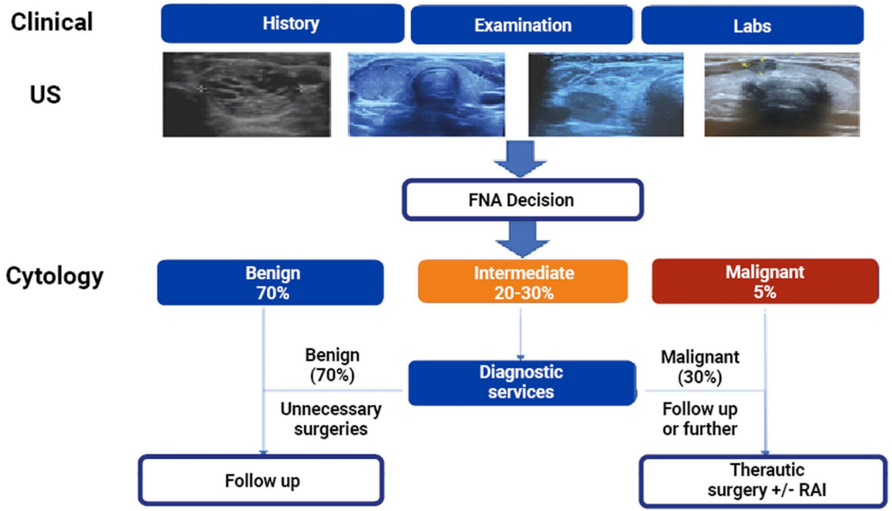
Gap of Knowledge

Figure 2

Decision Framework Under Current Standard of Care for Thyroid Nodule Evaluation



Note. Rajab, M., & Payne, R. J., 2025.

- Up to 30% of FNAs are indeterminate.
- Over 75% of “suspicious” surgeries reveal benign disease.
- TI-RADS and FNA alone cannot capture everything.

Gap of Knowledge

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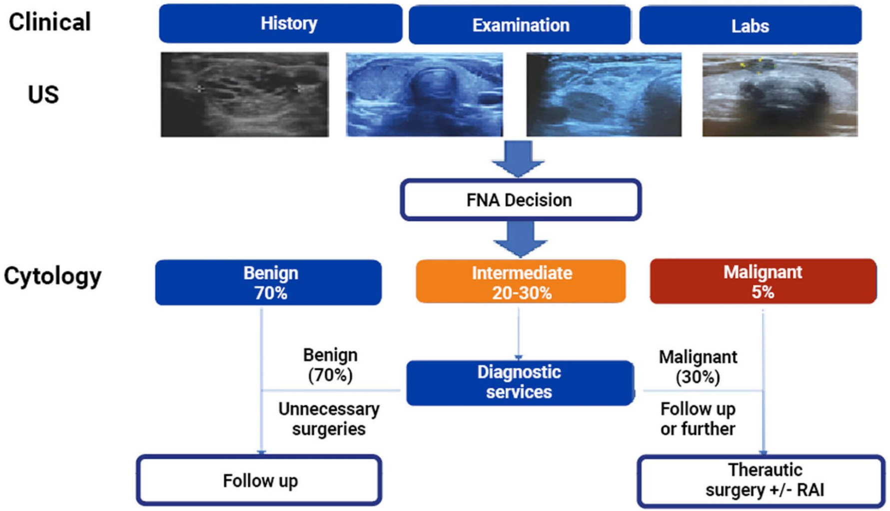

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- Up to 30% of FNAs are indeterminate.
- Over 75% of "suspicious" surgeries reveal benign disease.
- TI-RADS and FNA alone cannot capture everything.
- Need for a model that reduces unnecessary surgeries while improving risk prediction.

Rationale

Thyroid nodule FNAs are indeterminate and most "suspicious" surgeries turn out benign.

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→

A multimodal, longitudinal model would improve risk prediction and reduce unnecessary surgeries.

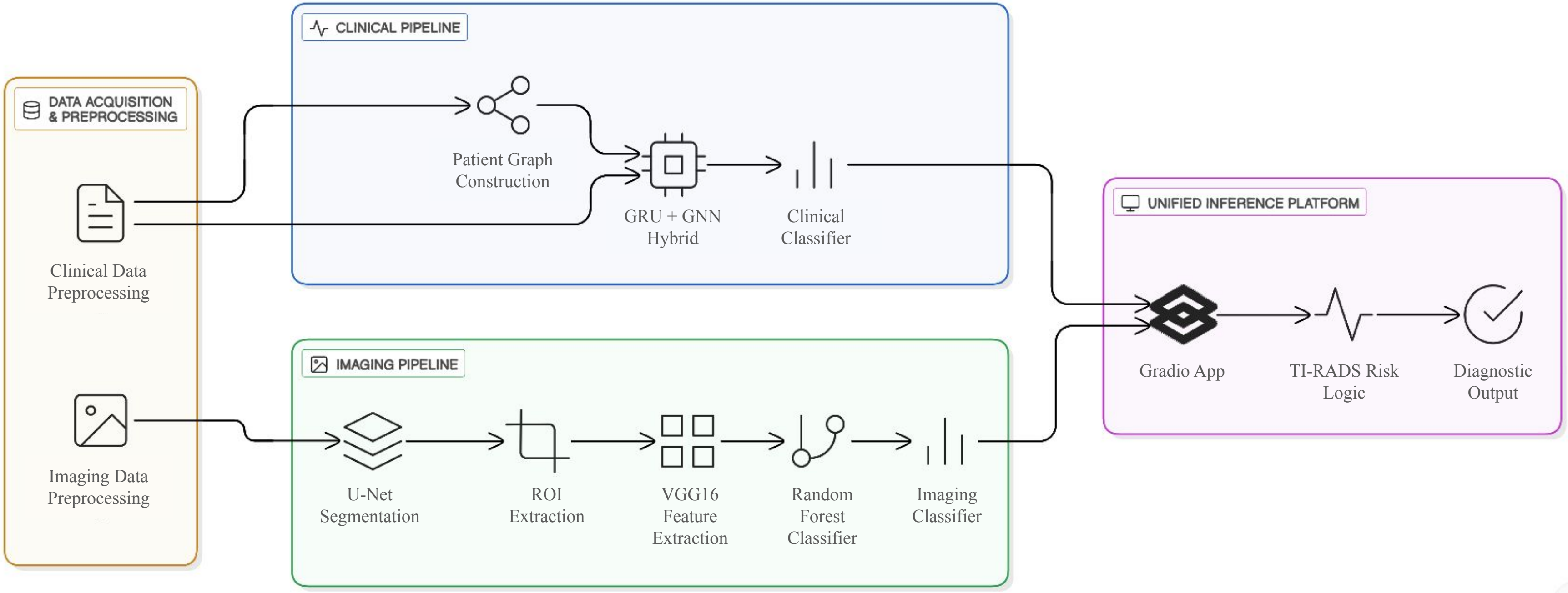
Research Question

Can a multimodal deep-learning framework integrating CNN prognostic markers, GRU temporal modeling, and GNN patient similarity improve thyroid nodule characterization and progression prediction beyond current clinical standards?

Methodology

Figure 3

End-to-End Diagnostic / Classification Methodology Pipeline

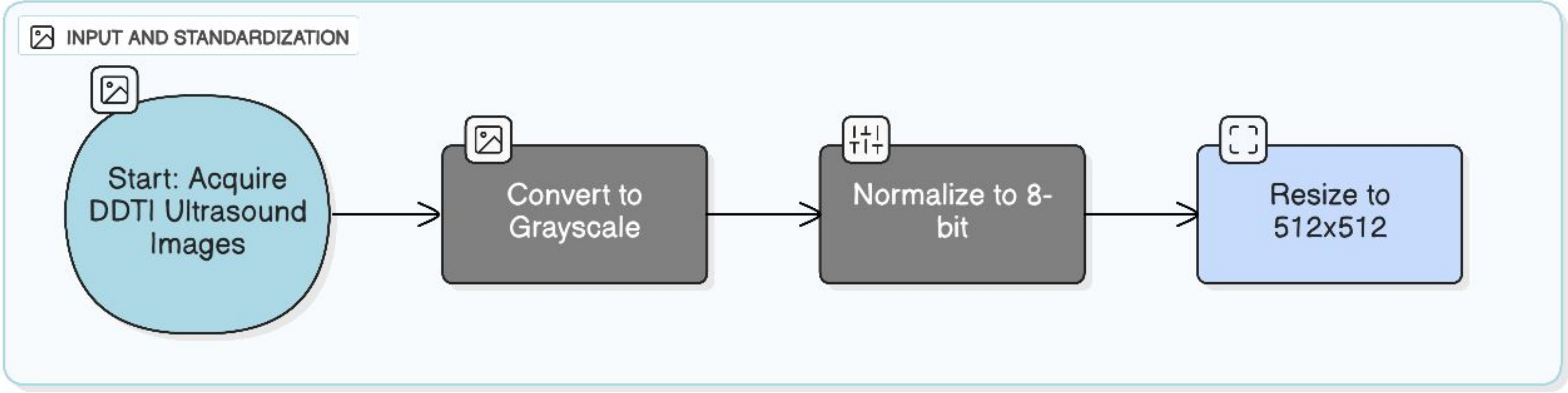


Note. Made by student researcher. Eraser.

Data Acquisition & Preprocessing

Figure 4

Input and Standardization Pipeline



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U-Net Segmented Mask Generation

Figure 5

Examples of U-Net Segmentation Masks

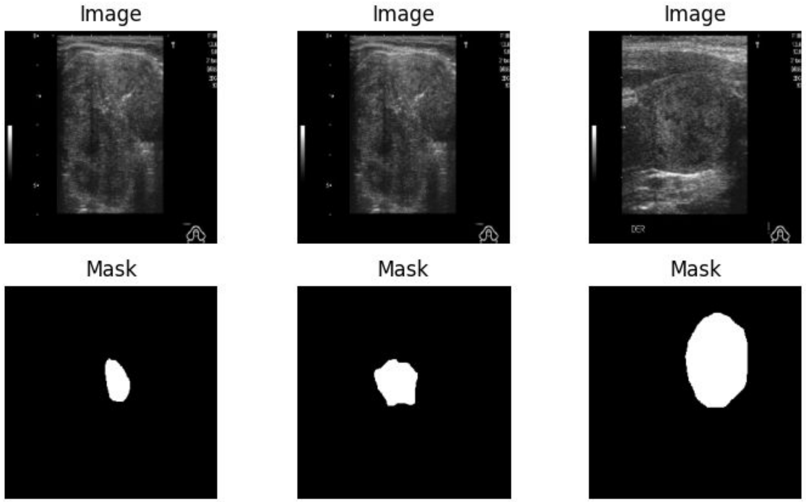
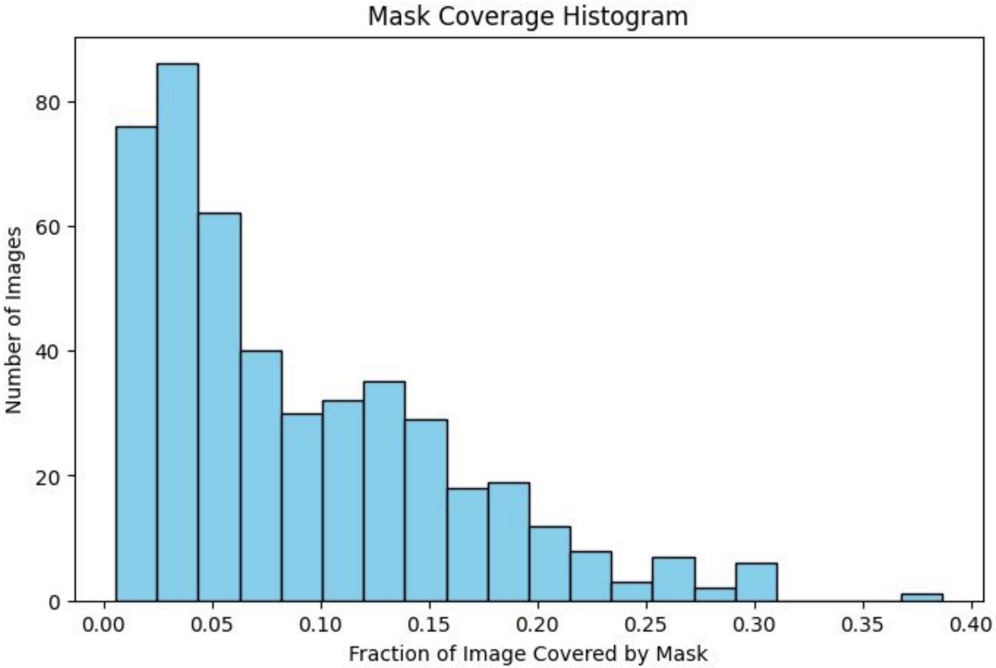


Figure 6

Distribution of Mask Coverage Area

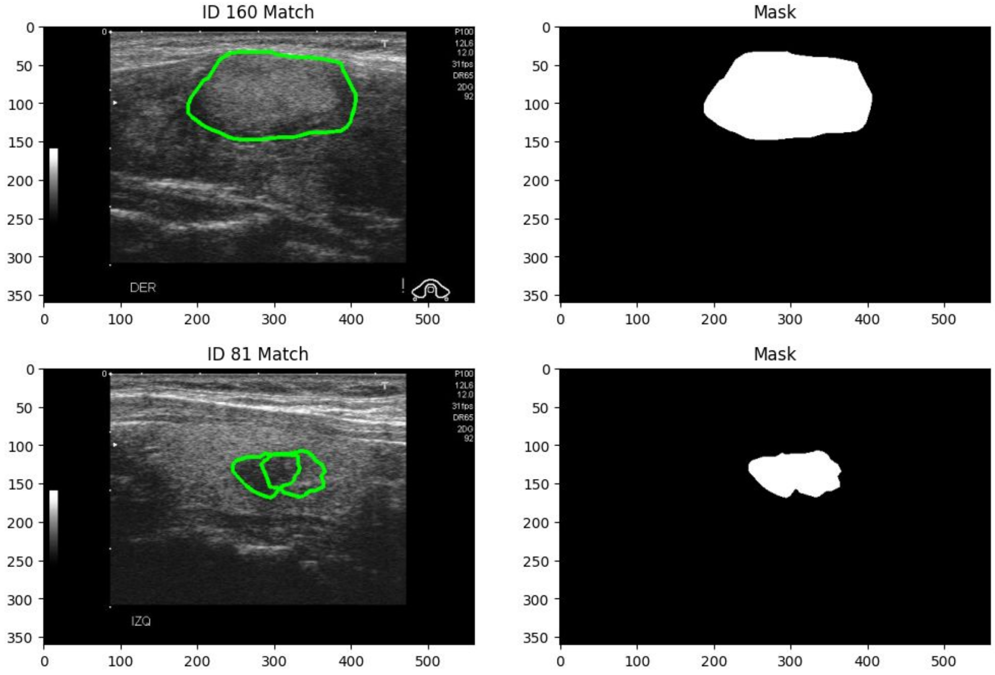


Note. Made by student researcher. Python, Lightning Studio.

U-Net Segmentation Interpretability

Figure 7

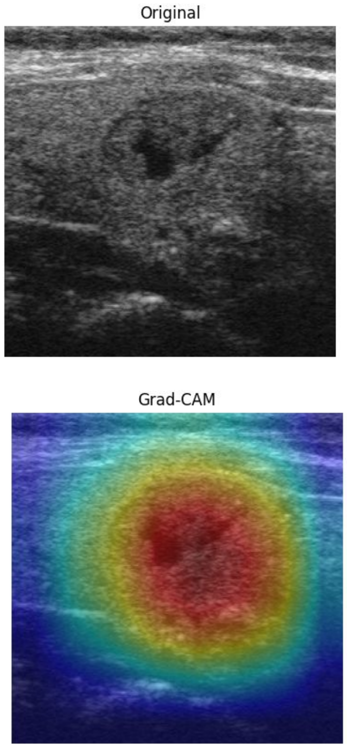
Ultrasound Mask Localizations of Thyroid Nodules



Note. Made by student researcher. Python, Lightning Studio.

Figure 8

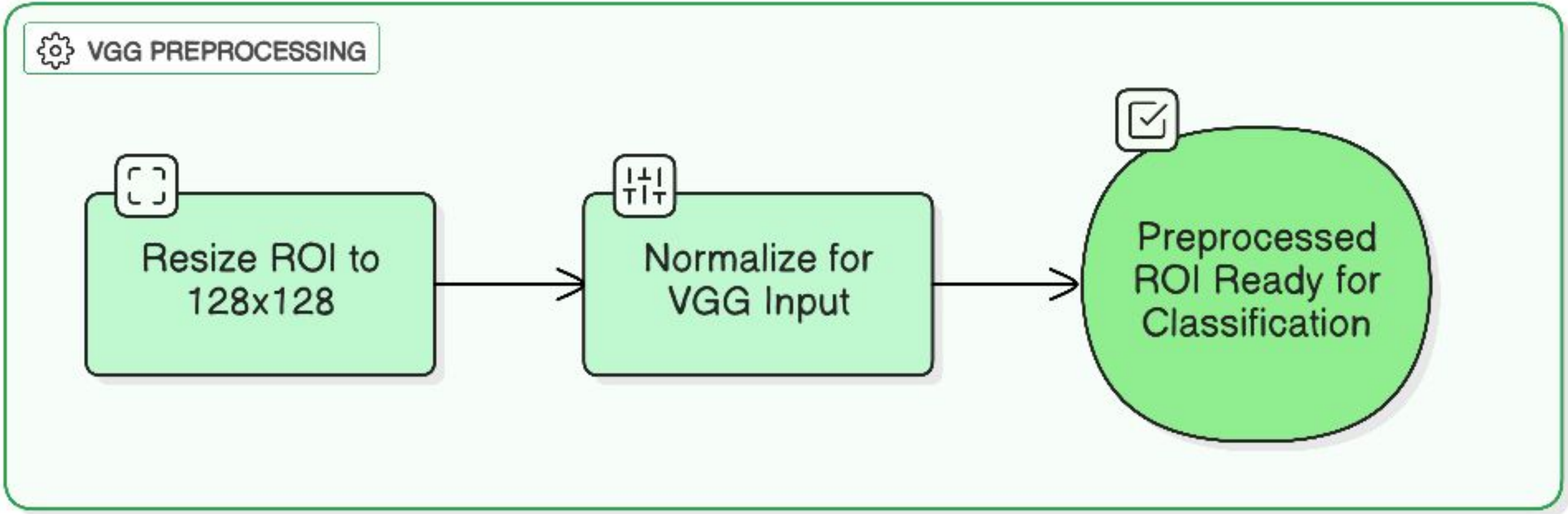
GradCAM Heatmaps



Data Acquisition & Preprocessing (Continued)

Figure 9

VGG Preprocessing Pipeline



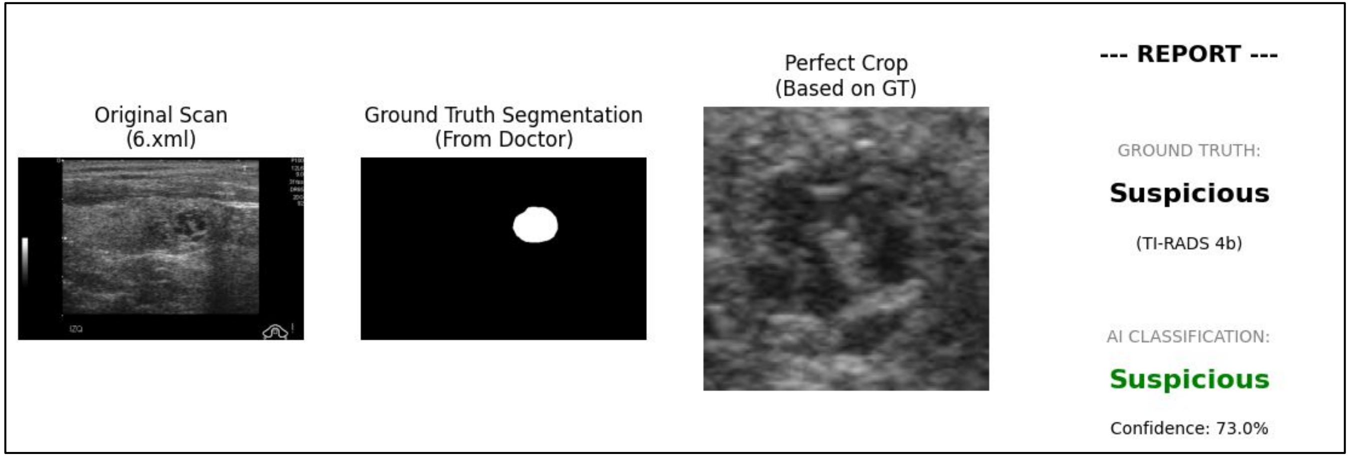
Note. Made by student researcher. Eraser.

CNN Classification

- Label mapping converts TI-RADS categories into 3 clinical classes

Figure 10

AI Diagnostic Workflow and Classification Result



Note. Made by student researcher. Python, Lightning Studio.

- VGG16-extracted imaging features
- RandomForest + SVM → Ensemble classifier models

Predictive GradCAM Heatmaps

Figure 11

Heatmap Pixel Occlusion Sensitivity

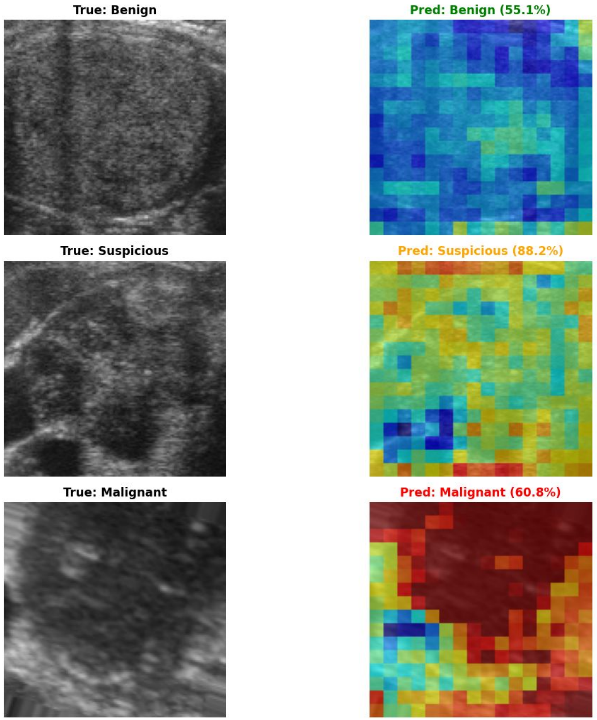
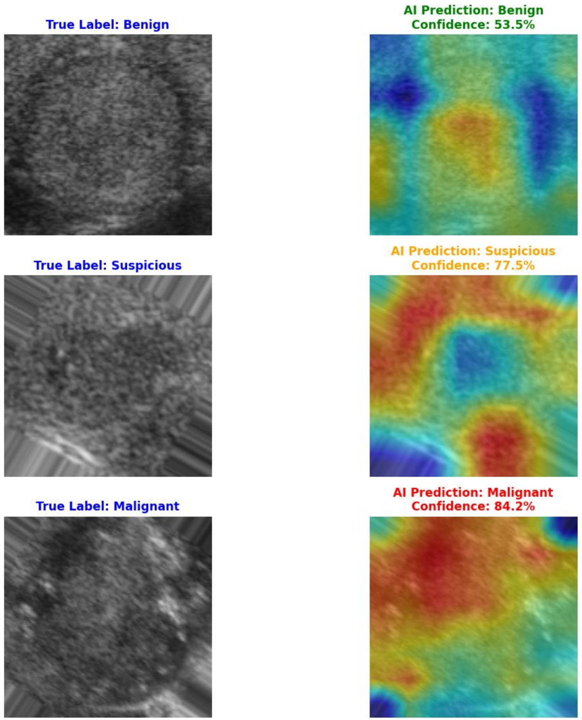


Figure 12

Heatmap Inter-Class Activation Maps



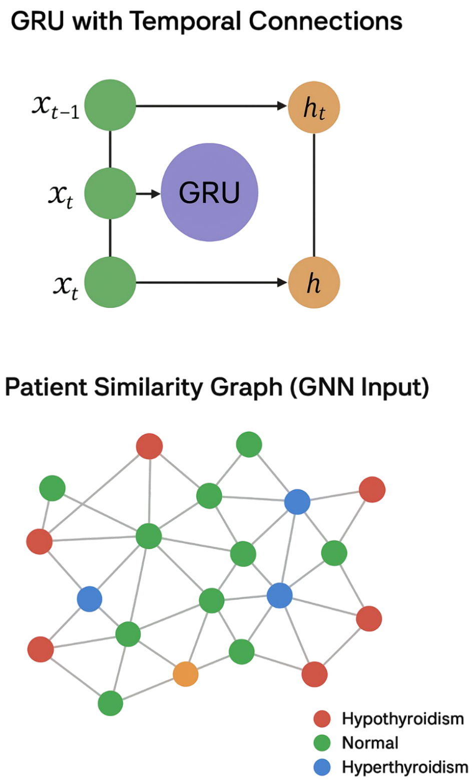
Note. Made by student researcher. Python.

Hybrid GRU–GNN Architecture

- GRU captures patient-specific feature representations
- GNN propagates information across similar patients
  - Nodes = patients
  - Edges = similarity (imaging, labs, demographics)
- Fusion layer combines temporal and relational embeddings

Figure 13

GRU and GNN Input Data Structures

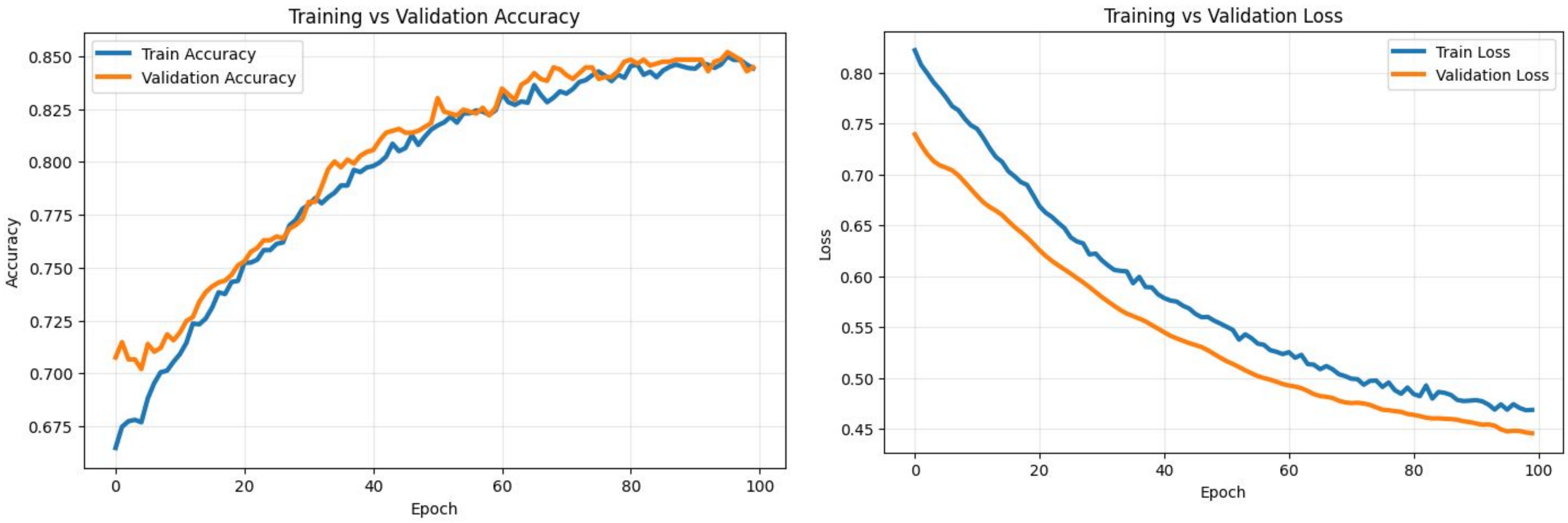


Note. Made by student researcher. Python.

GRU–GNN Training

Figure 13

Training vs. Validation Accuracy and Loss



Note. Made by student researcher. Python.

GRU–GNN Performance

Table 1

GRU-GNN Performance Metrics

	Precision	Recall	F1-Score	Support
Class 0	0.93	0.76	0.84	300
Class 1	0.86	0.99	0.92	258
Class 2	0.84	0.85	0.85	262
Class 3	0.80	0.84	0.82	282
Accuracy	—	—	0.86	1102
Macro Average	0.86	0.86	0.86	1102
Weighted Average	0.86	0.86	0.85	1102

Note. Made by student researcher.

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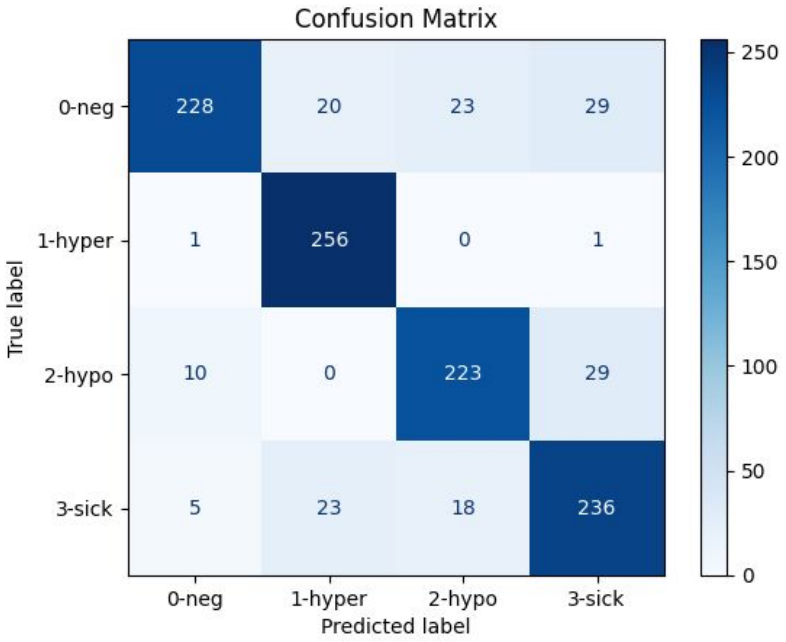
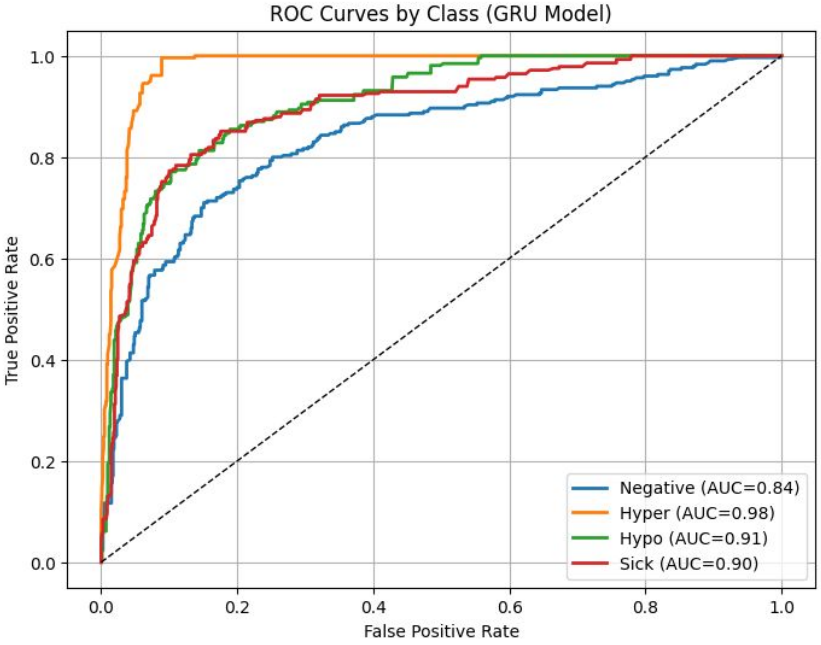
GRU–GNN Performance

Figure 14

Confusion Matrix & ROC Curve



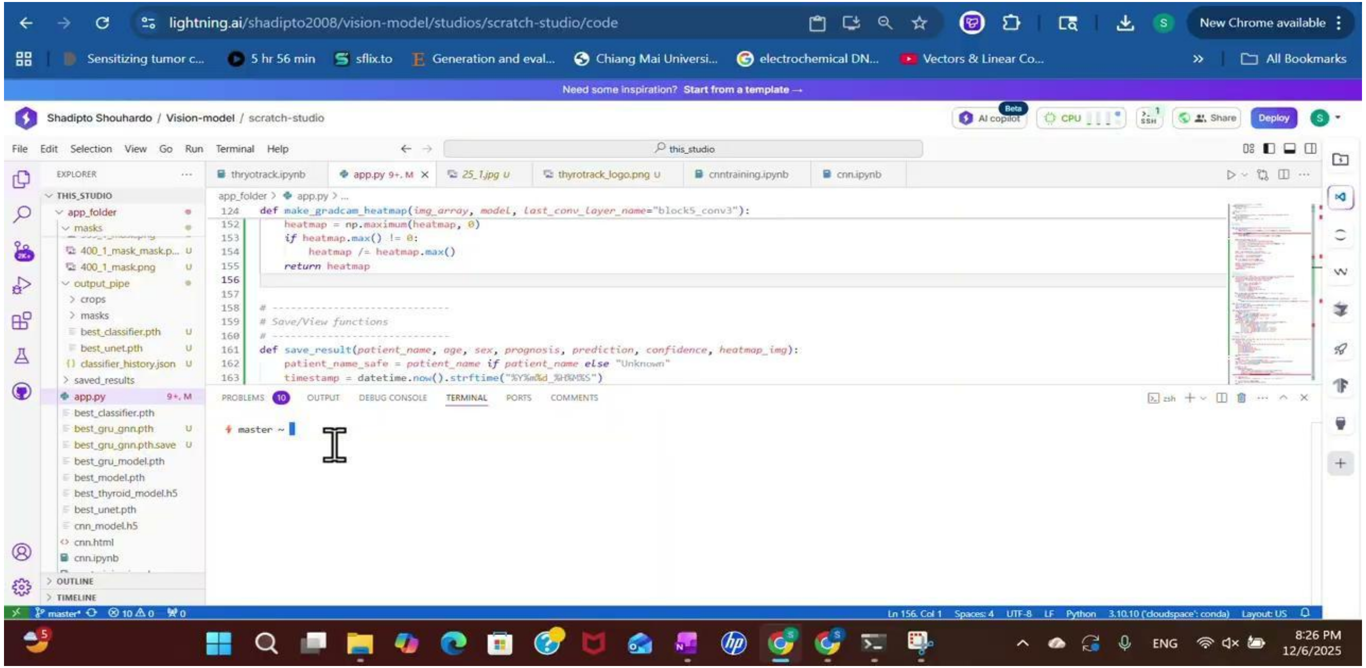


Note. Made by student researcher. Lightning Studio, Python.

Unified Inference Platform

Video 1

Gradio Frontend Interface Demo



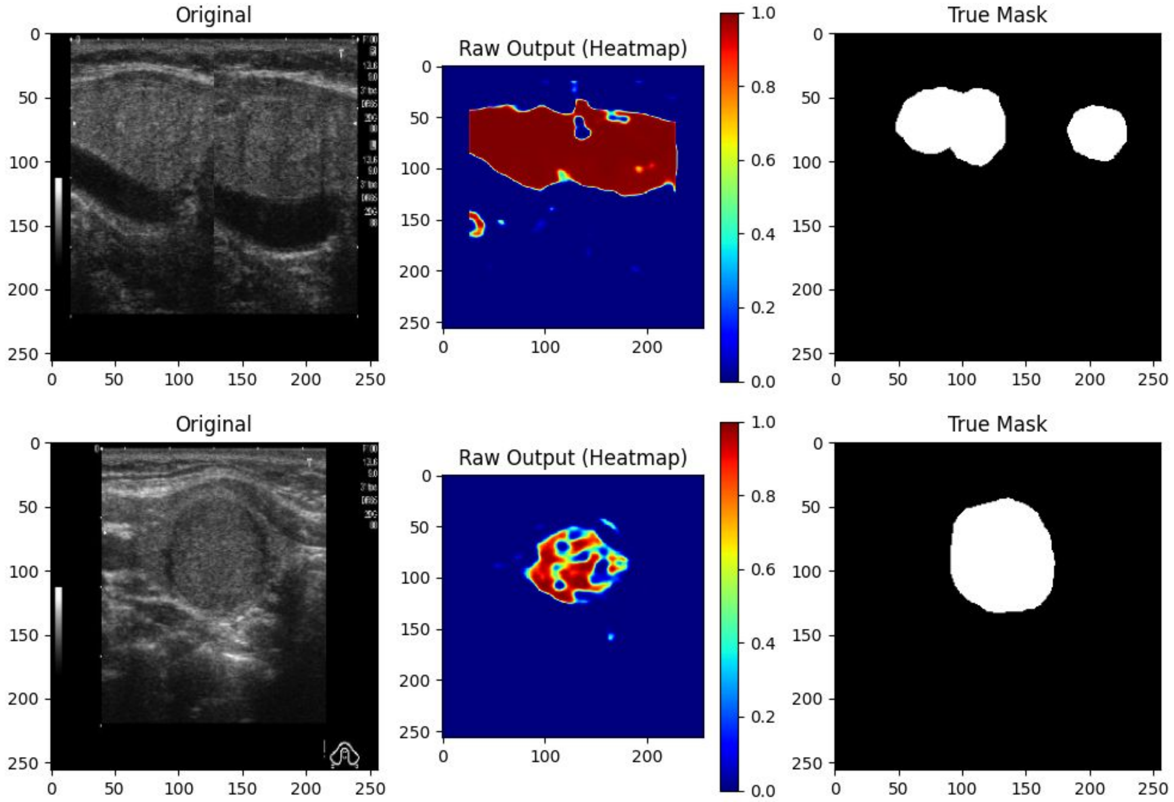
Note. Filmed by student researcher. Lightning Studio, Python.

Limitations

- Limited longitudinal depth
- Class imbalance
    - Malignant/high risk
- Single-dataset bias
- Residual gray-zone errors
    - Bethesda III–IV

Figure 15

Grad-CAM Underrepresentation Visualization



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Conclusion

- ThyroTrack provides a comprehensive platform for thyroid cancer prognosis and nodule assessment.
- Validated results demonstrate potential for multi-center application and longitudinal tracking.
- Ongoing enhancements aim to further optimize accuracy, dataset breadth, and real-world adoption. Integration potential into clinical workflows

References

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